

Machine Learning for Physicists

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Preface

So, you're a physicist looking to dive into the world of machine learning. You have a deep understanding of calculus, are familiar with experimental data-fitting methods, and have mastered concepts like entropy and non-linearity—yet you have only a vague idea of how a machine learning algorithm is actually written and how it works? Then this book is for you. And the good news is: your learning curve will grow fast. I've been in your shoes. I used to view machine learning as a distant horizon, but I soon realized that the knowledge I'd gained in physics made it incredibly easy to quickly master the concepts and engineering behind the field. So, come along and be amazed.

Chapter 1

Single-neuron ML

You have been using a single-neuron network since the first time you fitted experimental data, but perhaps no one ever told you that. Let's recall the basics. You have a set of experimental data points

$$(x_i, y_i) \tag{1.1}$$

through which to fit a linear function

$$f(x) = ax + b. \tag{1.2}$$

The Mean Squared Error, defined as

$$\begin{aligned} \text{MSE}(a, b) &= \frac{1}{N} \sum_{i=1}^N [y_i - f(x_i)]^2 \\ &= \frac{1}{N} \sum_{i=1}^N [y_i - (ax_i + b)]^2 \end{aligned} \tag{1.3}$$

tells us how distant the curve $f(x)$ is from the datapoints. Fitting $f(x)$ to the datapoints (x_i, y_i) means minimizing $\text{MSE}(a, b)$, that is, finding a and b such that

$$\frac{\partial}{\partial(a, b)} \text{MSE}(a, b) = 0. \tag{1.4}$$

No surprises so far, right? Well, what if I tell you the above is the simplest possible neural network? Indeed, it is a single-neuron network, where (1.2) is the activation function, (1.3) is the loss function and (1.4) is a very rudimentary backpropagation method.

Solving (1.4) for a and b , which, in this simple case is possible analytically, fixes the parameters to a_0 and b_0

$$\begin{aligned} a_0 &= \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \\ b_0 &= \bar{y} - a\bar{x} \end{aligned} \tag{1.5}$$

and we end up with a model $f_0(x) = a_0x + b_0$ (in (1.5), $\bar{x}$ and $\bar{y}$ are the mean values of the x_i and y_i datapoints). This model is a single neuron, no hidden layers: it receives one input parameter and outputs a number. f_0 has learned from data (1.1) through (1.4) and can now predict on any given x . Pretty cool, huh? The concepts were sitting there all the time. Take a moment to re-read the definitions and let's jump to the second simplest model: a two-neuron network.

Chapter 2

Two-neuron ML